656342

Course Name: Discrete Mathematics

Course Outcome

CO1-Understand the notion of mathematical thinking and proofs to solve the problem.
CO2-Apply the basics of discrete probability and number theory to solve the real world problem.
CO3-Analyze basic discrete structures and algorithms using effectively algebraic techniques.
CO4-Analyze mathematical concepts like sets, reasoning, relational algebra and graph theory to solve optimization problems.
CO5-Analyze the validity of an argument using logical notation.
CO6. Demonstrate the basic structures of proof techniques to write and evaluate the validity of arguments.
CO7. Understand the basic principles of sets, set equalities and operations in sets.
CO8. Apply counting principles to determine probabilities.

Printed Pages: 2 **University Roll No. .....**

Mid Term Examination, Even Semester 2024-25
B. Tech. (AIML&IIoT/BCT/CCV/DA/CSF), IVTh Sem, IInd Year
BCSC 1010, Discrete Mathematics

Time: 2 Hours **Maximum Marks: 30**

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Prove That $A-(B\cup C)=(A-B)\cap(A-C)$	3	CO4	E	P
2	(I) How many ways are possible to distribute 7 different candies to 3 people where each gets only 1 candy? (II) In how many ways can the letters of the word 'POWER' be arranged? (III) How many six-digit numbers can be formed with digits 2,3,5, 6, 7, and 9 and with distinct digits?	3	CO4	E	P
3	Prove that the relation R is an equivalence type in the set P= { 3, 4, 5,6 } given by the relation R = { (p, q):|p-q| is even }.	3	CO7	E	P
4	(I) A drawer contains 12 red and 12 blue socks, all unmatched. A person takes socks out at random in the dark. How many socks must he take out to be sure that he has at least two blue socks? (II) How many ways a 5-member committee can be formed out of 10 people if two particular people must be included?	3	CO8	An	C
5	By using mathematical induction prove that $6^{n+2}+7^{2n+1}$ is divisible by 43 for each positive integer n	3	CO1	C	C

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Solve the Recurrence Relation using Generating function $a_n - 6a_{n-1} + 9a_{n-2} = 0$ $n \geq 2$ $a_0 = 1$, $a_1 = 9$	5	CO8	E	P
7	Answer these questions for the POSET ({1,2,3,5,6,10,15,30}, /) (a) Find the maximal elements. (b) Find the minimal elements. (c) Is there a greatest element? (d) Is there a least element? (e) find all upper bounds of {6,10}.	5	CO1	E	P
8	In a city of 1000 families, it was found that 40% families buy newspaper A, 20 families buy newspaper B, and 10% buy C. Only 5% families buy A and B, 3% buy B and C and 4% buy A and C and 2% families buy all the three newspapers. Find the number of families which buy: (i) A only (ii) B only (iii) None of A, B and C.	5	CO4	E	P